distribution over the rotor disk for cases 1A and 1, respectively. Case 1A uses uniform inflow and case 1 the nonuniform inflow; neither includes stall aerodynamics or the effect of the blade motion due to stall.

Table 1 presents the performance and trim data calculated for the seven cases considered. Cases 1, 1A, and 1B are with no stall; cases 1A and 1B examine the influence of uniform inflow and rigid blade motion, respectively. Case 2 uses static stall. Compared with the no-stall and the dynamic-stall cases, the static stall case shows very high values for power required, rotor flapping, and control angles (collective and cyclic), even though the thrust is still somewhat below the target value. These results imply that this operating condition (the combination of thrust, propulsive force, and forward speed) would be beyond the capability of this rotor if the blade stalled statically. The loads and the bending and torsion moments calculated for case 2 are unrealistic, therefore, and are not included in the other comparisons presented here. The three dynamic stall methods (cases 3, 4, and 5) give essentially the same results for performance and trim. Comparing, for example, the power required for these three cases with that for the no-stall case, it is evident that the stall effects are quite important. However, the effects predicted using the dynamic-stall models are much less than those obtained using static-stall data (case 2).

Table 1 also presents the results for the vertical and inplane root shear forces of the rotor. Only the harmonics transmitted to the helicopter body are given. There is a trend to increased magnitude of the higher harmonics due to stall. The variation of the root shear forces between the cases is an indication of the general sensitivity of the problem of calculating helicopter rotor vibration, as well as the difference between the stall models.

Figure 4 compares the elastic torsion motion of the blade, for the three dynamic stall methods. Since only one mode is used to represent the torsion motion, the torsion moments are directly proportional to the torsion deflection. Figure 4 shows significant differences in the details of the torsion moment waveform, particularly on the retreating side where dynamic stall is involved. The peak-to-peak torsion moments given by the three methods differ by about 40 percent. Figures 5 through 8 show the peak-to-peak and mean flatwise and edgewise blade bending moments for the three dynamic stall cases and the no-stall results (case 1). The mean bending moments predicted by the three stall cases are roughly the same. The higher mean edgewise moment compared with the no-stall case is due to the increased blade section drag associated with stall. Consequently the increase in the edgewise bending moment is about the same as the increase in the power required. The calculated peak-to-peak bending moments show roughly the same general characteristics, but the magnitude predicted by the three stall methods differ by about 25 to 40 percent.

The differences in the calculated blade torsion and bending moments may be traced to the blade aerodynamic loading. Figures 9 to 12 present the section lift force, and figures 13 to 16 the section pitching moment for the no-stall case (case 1) and the three dynamic-stall methods (cases 3, 4, and 5). There are significant differences in the aerodynamic loading, particularly in the pitch moment, due to the fundamental differences in the calculation of the dynamic stall loads in the three methods.